# INDIAN SPACE OLYMPIAD

# 2026 MOCK QUESTION PAPERS

# JUNIOR LEVEL

## CLASSES V-VI

### 5 Mock Papers | 50 Questions Each | 250 Questions

Strictly aligned to the ISO Junior Level syllabus for Classes V-VI.Questions 41-50 in every paper form the Olympiad Challenge.

**Indian Space School**

# How to Use These Mock Papers

These five papers are written specifically for students of Classes V and VI. The questions remain within the supplied Junior Level syllabus. The language and calculations are age appropriate, while the final ten questions of each paper provide a stronger Olympiad challenge.

Each paper contains 50 single-correct multiple-choice questions. Questions 1-40 test core knowledge, observation and application. Questions 41-50 require deeper reasoning, short calculations, interpretation or careful comparison.

Suggested time: 60 minutes per paper. Suggested marking: 1 mark for every correct answer and no negative marking during practice. Students should attempt a paper independently before checking the answer key.

## Syllabus Coverage

| Unit | Coverage |
|---|---|
| Fundamentals of Astronomy | night sky, directions, constellations, celestial sphere, telescopes, observations and events |
| The Solar System | Solar System formation, Sun, planets, dwarf planets, moons, asteroids, comets and meteors |
| Stars, Galaxies and the Universe | star properties, constellations, clusters, nebulae, galaxies and the Universe |
| Earth and Moon | Earth structure and atmosphere, rotation, revolution, seasons, time zones, Moon phases, eclipses and tides |
| Satellites and Space Technology | natural and artificial satellites, communication, navigation, weather, remote sensing and applications |

| | |
|---|---|
| | |
| Rockets and Launch Vehicles | rocket principles, stages, payloads, launch sites and reusable rockets |
| ISRO and India's Space Missions | ISRO, major centres, PSLV, GSLV, Chandrayaan, Mangalyaan, Aditya-L1 and Gaganyaan |
| Space Exploration | astronauts, space stations, Moon and Mars missions, space agencies and famous missions |
| Basic Physics for Space Science | force, motion, gravity, Newton's Laws, work, energy, heat, light, sound, electricity and magnetism |
| Scientific Reasoning and Logical Aptitude | observation, patterns, data interpretation, analytical thinking and numerical aptitude |
| General Space Awareness | current missions, discoveries, scientists, astronauts, events and applications of space technology |

## Instructions for Students

Choose the best answer from A, B, C and D.

Use rough work for calculations.

Read data and sequence questions carefully.

Do not spend too long on one question; return to it later.

Attempt all ten Olympiad Challenge questions.

## Information Cut-off

Mission-status statements were checked against official ISRO and NASA information available through 23 July 2026. Completed missions, operational missions, missions under testing and future missions are described accordingly.

# MOCK PAPER 1

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes     Maximum Marks: 50

*Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question.*

**1. Astronomy is the study of**

A. objects and events in space     B. plants in a garden     C. only rocks under Earth     D. languages spoken by people

**2. Which object is usually easiest to see in the daytime sky?**

A. the Milky Way     B. the Sun     C. a distant galaxy     D. a faint constellation

**3. A group of stars that seems to form a pattern is called a**

A. satellite     B. meteorite     C. constellation     D. launch pad

**4. Which instrument helps us see faint and distant objects in the sky?**

A. microscope     B. thermometer     C. compass only     D. telescope

**5. The Solar System formed from a large cloud of gas and dust. This is the basic idea of the**

A. solar nebula theory     B. shadow theory     C. tide theory     D. rain cycle

**6. The Sun is a**

A. planet     B. star     C. moon     D. comet

**7. Which planet is closest to the Sun?**

A. Earth     B. Mars     C. Mercury     D. Jupiter

**8. Which of these is a dwarf planet?**

A. Venus     B. Saturn     C. Earth     D. Pluto

**9. A space rock that reaches the ground is called a**

A. meteorite     B. meteor     C. meteoroid only     D. comet tail

**10. Stars appear bright mainly because they**

A. reflect only moonlight     B. give out their own light     C. are made of ice     D. are close to every observer

**11. A large group of stars held together by gravity is a**

A. time zone     B. rocket stage     C. star cluster     D. solar panel

**12. A cloud of gas and dust in space is called a**

A. crater     B. constellation map     C. telescope     D. nebula

**13. Our Solar System is inside the**

A. Milky Way galaxy     B. Andromeda Moon     C. Kuiper planet     D. Earth cluster

**14. Earth's outermost solid layer is the**

A. mantle     B. crust     C. outer core     D. inner core

**15. Most weather takes place in the lowest layer of the atmosphere called the**

A. exosphere     B. mantle     C. troposphere     D. magnetosphere

**16. Day and night are mainly caused by**

A. Earth's revolution only     B. the Moon changing shape     C. clouds covering the Sun     D. Earth's rotation

**17. The Moon looks fully lit from Earth during the**

A. full Moon B. new Moon C. solar eclipse D. first sunrise

**18. Which is a natural satellite?**

A. INSAT B. the Moon C. a weather balloon D. a rocket

**19. A communication satellite mainly helps to**

A. make oceans salty B. create gravity C. send signals over long distances D. stop Earth rotating

**20. A weather satellite helps scientists observe**

A. the inside of the Sun only B. underground minerals only C. the sound of stars D. clouds and storms

**21. Remote sensing means collecting information about an area**

A. from a distance using instruments B. only by touching it C. only by digging D. without using any data

**22. A rocket moves upward because hot gases are pushed**

A. upward B. downward C. sideways only D. nowhere

**23. A launch vehicle is built in stages mainly so that it can**

A. carry no fuel B. remain on the launch pad C. drop empty parts and become lighter D. avoid using engines

**24. The useful object carried by a rocket is called the**

A. shadow B. crater C. constellation D. payload

**25. India's major space launch centre at Sriharikota is the**

A. Satish Dhawan Space Centre B. Vikram Sarabhai Library C. Aryabhata Observatory on Mars D. Indian Ocean Station

**26. ISRO stands for**

A. International Satellite Repair Office B. Indian Space Research Organisation C. Indian Science Rocket Orbit D. Interplanetary Space Radio Organisation

**27. PSLV is best known as an Indian**

A. space station B. planet C. launch vehicle D. astronaut suit

**28. Chandrayaan missions explore the**

A. Sun B. Jupiter C. Milky Way centre D. Moon

**29. Mangalyaan was sent to study**

A. Mars B. Venus C. Mercury D. Saturn

**30. A person trained to travel and work in space is an**

A. astronomer only B. astronaut C. architect D. athlete

**31. The International Space Station travels around**

A. the Sun only B. Mars C. Earth D. the Moon

**32. NASA is the space agency of the**

A. Moon B. European Union only C. planet Mars D. United States

**33. A push or a pull is called a**

A. force B. shadow C. orbit D. temperature

**34. The force that pulls objects towards Earth is**

A. magnetism only B. gravity C. light D. sound

**35. Work is done when a force moves an object through a**

A. colour B. name C. distance D. temperature only

**36. Which object allows light to pass through clearly?**

A. wooden board B. brick wall C. metal sheet D. clear glass

**37. Complete the pattern: Sun, Moon, Sun, Moon, Sun, __.**

A. Moon B. Mars C. rocket D. star cluster

**38. Which one does not belong with the others?**

A. Mercury B. rocket C. Venus D. Earth

**39. Aditya-L1 studies the**

A. Moon B. Mars C. Sun D. rings of Saturn only

**40. Which daily service commonly uses navigation satellites?**

A. boiling water B. growing hair C. making shadows disappear D. finding a route on a phone map

## OLYMPIAD CHALLENGE - QUESTIONS 41-50

*These ten questions are deliberately more demanding. Read carefully, reason step by step and calculate where needed.*

**41. A student faces east at sunrise. Which direction is directly behind the student?**

A. west B. north C. south D. up

**42. A planet is between Venus and Mars in distance from the Sun. Which planet is it?**

A. Mercury B. Earth C. Jupiter D. Saturn

**43. A bright object gives out its own light and is much larger than Earth. It is most likely a**

A. planet B. moon C. star D. asteroid

**44. At 6 p.m. in one place, a city one time zone to the east is usually at about**

A. 5 p.m. B. 6 a.m . C. 12 noon D. 7 p.m.

**45. A farmer wants a satellite image showing which fields are green. Which technology is most useful?**

A. remote sensing B. sound recording C. a launch pad D. a star chart

**46. Stage 1 of a rocket has finished burning fuel. Why is it released?**

A. to increase air resistance B. to reduce mass for the remaining flight C. to stop all engines D. to pull the rocket back to Earth

**47. Which mission-object pair is correctly matched?**

A. Mangalyaan - Sun B. Aditya-L1 - Mars C. Chandrayaan - Moon D. Gaganyaan - Jupiter

**48. Astronauts in an orbiting space station seem to float mainly because they and the station are**

A. outside all gravity B. lighter than air C. pushed up by sunlight D. falling around Earth together

**49. A 5 N force moves a toy 4 m in the same direction. The work done is**

A. 20 J B. 1.25 J C. 9 J D. 40 J

**50. A class records stars seen on four nights: 8, 12, 12 and 15. On which two nights was the number the same?**

A. first and second B. second and third C. first and fourth D. third and fourth

# MOCK PAPER 2

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes     Maximum Marks: 50

*Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question.*

**1. The imaginary dome on which sky objects appear to lie is called the**

A. ocean floor     B. celestial sphere     C. rocket chamber     D. planetary crust

**2. The direction halfway between north and east is**

A. south-west     B. north-west     C. north-east     D. south-east

**3. A notebook used during sky watching should record the time, place and**

A. price of the pencil     B. student height only     C. colour of the cover     D. direction of the object

**4. A solar eclipse is an example of an**

A. astronomical event     B. ocean current     C. soil layer     D. simple machine

**5. Which planet is famous for its large rings?**

A. Mercury     B. Saturn     C. Venus     D. Mars

**6. Which group contains only giant planets?**

A. Mercury, Venus, Earth and Mars     B. Earth, Mars, Jupiter and Saturn     C. Jupiter, Saturn, Uranus and Neptune     D. Venus, Earth, Uranus and Neptune

**7. The Sun provides Earth mainly with**

A. soil and rocks     B. tides only     C. oxygen from its surface     D. light and heat

**8. A comet is mainly made of ice, dust and**

A. rocky material     B. liquid metal only     C. wood     D. clouds from Earth

**9. A meteoroid that glows while moving through the atmosphere is seen as a**

A. planet     B. meteor     C. moon     D. galaxy

**10. A star can look brighter than another because it may be larger, hotter or**

A. made of soil     B. inside the Moon     C. closer to Earth     D. moving slower than sound

**11. A constellation is a pattern of stars seen from**

A. inside the Sun     B. under the ocean     C. inside a rocket engine     D. Earth

**12. The Milky Way contains**

A. billions of stars     B. only eight planets     C. one single star     D. no gas or dust

**13. The Universe includes**

A. only Earth and Moon     B. galaxies, stars, planets and space     C. only the Solar System     D. only objects we can touch

**14. Earth's thick layer below the crust is the**

A. troposphere     B. inner core only     C. mantle     D. ocean

**15. Earth takes about one year to complete one**

A. rotation on its axis     B. Moon phase     C. solar eclipse     D. revolution around the Sun

**16. Seasons occur mainly because Earth is tilted and**

A. revolves around the Sun B. stops rotating C. moves closer to the Moon every month D. changes its mass

**17. During a lunar eclipse, which object is between the Sun and Moon?**

A. Mercury B. Earth C. a comet D. the space station

**18. A navigation satellite helps a traveller find**

A. the age of a star B. the taste of water C. location and direction D. the weight of sunlight

**19. A satellite used for television programmes is mainly a**

A. natural satellite B. meteorological balloon only C. launch vehicle D. communication satellite

**20. Satellite images can help during floods by showing**

A. which areas are covered by water B. the sound of rain C. the inside of every house D. the names of all people

**21. Solar panels on a satellite change sunlight into**

A. sound energy only B. electrical energy C. soil D. gravity

**22. The place from which a rocket is sent into space is a**

A. galaxy B. crater C. launch site D. constellation

**23. Rocket fuel and an oxidiser react in the engine to produce hot**

A. ice B. sand C. moonlight D. gases

**24. A reusable rocket is designed so that some part can**

A. fly again after recovery B. turn into a satellite C. remain in space forever D. avoid all maintenance

**25. The part at the top of a launch vehicle may protect the payload with a**

A. crater B. fairing C. compass D. tide

**26. The Vikram Sarabhai Space Centre is closely connected with**

A. ocean fishing B. weather on Jupiter only C. launch vehicle development D. building railway engines

**27. GSLV is designed to launch satellites, including some into**

A. underground tunnels B. the ocean floor C. inside the Sun D. high Earth orbits

**28. Aditya-L1 observes the Sun from near the Sun-Earth**

A. L1 point B. north pole C. asteroid belt D. Moon crater

**29. Gaganyaan is India's programme for**

A. underwater farming B. human spaceflight C. building a Moon-sized telescope D. moving Earth

**30. A space station is a place where astronauts can**

A. stand on the Sun B. stop all satellites C. live and carry out experiments in orbit D. create planets

**31. The European Space Agency is commonly shortened to**

A. ESO only B. EGA C. EVA D. ESA

**32. A rover is a vehicle designed to**

A. move across the surface of another world B. orbit only without landing C. measure sound in a classroom D. launch itself without a rocket

**33. Newton's first law is also called the law of**

A. reflection B. inertia C. tides D. heat

**34. When the same force acts on a lighter object, it usually produces**

A. less motion in every case B. no change at all C. greater acceleration D. a lower temperature only

**35. Sound needs a material medium such as air, but light can travel through**

A. only water B. only metal C. nothing at all D. vacuum

**36. A magnet has two ends called**

A. poles B. orbits C. stages D. phases

**37. Which sequence is in increasing order?**

A. 8, 6, 4, 2 B. 2, 4, 6, 8 C. 2, 6, 4, 8 D. 4, 2, 8, 6

**38. A chart shows 3 rockets on Monday, 5 on Tuesday and 4 on Wednesday. The highest number is on**

A. Monday B. Wednesday C. Tuesday D. all three days

**39. Chandrayaan-3 demonstrated a safe soft landing on the**

A. Sun B. Mars C. Venus D. Moon

**40. Which technology helps television channels send programmes across large regions?**

A. communication satellites B. meteorites C. constellations D. tides

## OLYMPIAD CHALLENGE - QUESTIONS 41-50

*These ten questions are deliberately more demanding. Read carefully, reason step by step and calculate where needed.*

**41. A star is seen low in the western sky after sunset. Which observation is most complete?**

A. only the star colour B. time, place, western direction and height above horizon C. only the observer name D. only the date

**42. A body is too small to be a planet, orbits the Sun beyond Neptune, and is round. It is most likely a**

A. natural satellite of Earth B. star cluster C. dwarf planet D. rocket stage

**43. Three objects are listed: a nebula, a star cluster and a galaxy. Which is largest in general?**

A. nebula always B. star cluster always C. all must be exactly equal D. galaxy

**44. It is summer in the Northern Hemisphere. Which statement is most likely true?**

A. the Northern Hemisphere is tilted more towards the Sun B. Earth is much closer to the Sun only C. the Moon blocks winter D. Earth has stopped rotating

**45. A satellite must photograph the same farming region every few days. Which ability is most important?**

A. producing rocket thrust B. repeated Earth observation from orbit C. making a comet tail D. changing time zones

**46. A two-stage rocket has used all fuel in stage one. Keeping the empty stage attached would mainly**

A. create extra useful fuel B. remove gravity C. make later acceleration harder D. increase payload mass without cost

**47. Which statement correctly compares PSLV and GSLV?**

A. PSLV is a planet and GSLV is a star B. both are space stations C. GSLV studies only oceans D. both are Indian launch vehicles used for different mission needs

**48. Why do astronauts exercise regularly on a space station?**

A. to reduce loss of muscle and bone strength B. to create gravity for Earth C. to recharge the Sun D. to move the station by running

**49. A 12 N force moves a box 3 m. How much work is done when force and motion are in the same direction?**

A. 4 J B. 36 J C. 15 J D. 144 J

**50. A pattern follows: 3, 6, 12, 24, __. What is next?**

A. 27 B. 30 C. 48 D. 36

# MOCK PAPER 3

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes     Maximum Marks: 50

*Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question.*

**1. At night, stars appear to move from east to west mainly because**

A. stars circle Earth once each minute    B. the Sun pushes stars    C. Earth rotates from west to east    D. the Moon turns the sky

**2. A compass is useful in sky observation because it helps find**

A. star temperatures directly    B. planet sizes    C. rocket fuel    D. directions

**3. Which event can be safely watched only with proper solar protection?**

A. a solar eclipse    B. a full Moon    C. a meteor shower    D. a constellation

**4. A telescope with a larger opening usually collects**

A. less light    B. more light    C. only sound    D. no information

**5. Earth is the __ planet from the Sun.**

A. first    B. fifth    C. third    D. eighth

**6. Mars is often called the Red Planet because of iron-rich material in its**

A. clouds of water only    B. rings    C. comet tail    D. surface rocks and dust

**7. Which planet is the largest?**

A. Jupiter    B. Earth    C. Mercury    D. Mars

**8. Most asteroids in our Solar System orbit between**

A. Earth and Moon    B. Mars and Jupiter    C. Sun and Mercury    D. Uranus and Neptune only

**9. A comet develops a visible tail when it comes closer to the**

A. Earth only    B. Milky Way centre    C. Sun    D. north pole

**10. Stars differ in colour partly because they have different**

A. numbers of moons    B. time zones    C. rocket stages    D. surface temperatures

**11. A blue-white star is generally hotter than a**

A. red star    B. blue star    C. white star    D. star with no colour

**12. Many young stars can form inside a**

A. meteorite    B. nebula    C. satellite dish    D. launch pad

**13. A galaxy is held together mainly by**

A. sound    B. air pressure    C. gravity    D. sunlight only

**14. Earth's centre contains the**

A. crust only    B. atmosphere    C. ocean floor only    D. core

**15. The ozone layer helps absorb much of the Sun’s harmful**

A. ultraviolet radiation    B. radio sound    C. gravity    D. visible green light only

**16. Earth completes one rotation in about**

A. 7 days    B. 24 hours    C. 30 days    D. 365 days

**17. The Moon phase that comes after new Moon begins as a**

A. waning crescent B. full Moon C. waxing crescent D. lunar eclipse

**18. A satellite orbiting Earth follows a curved path because of its motion and**

A. sound waves B. clouds C. magnetic paint D. gravity

**19. Navigation satellites are used by ships, aircraft and phones to find**

A. position B. star colour C. rocket temperature only D. Moon age

**20. Remote sensing can help map**

A. only constellations B. forests, farms and water bodies C. only rocket engines D. only sounds

**21. A spacecraft needs antennas mainly to**

A. make craters B. produce tides C. communicate with Earth D. form stars

**22. The upward push that moves a rocket is called**

A. tide B. shadow C. orbit D. thrust

**23. The first rocket stage usually operates**

A. soon after lift-off B. after the payload lands C. only on the Moon D. before fuel is added

**24. A payload fairing protects a satellite mainly during**

A. its entire life under the ocean B. flight through the atmosphere C. a lunar eclipse D. a star forming

**25. A reusable launch system can lower costs by**

A. making every rocket heavier B. removing all testing C. using recovered parts again D. avoiding navigation

**26. The main headquarters of ISRO is in**

A. New York B. Tokyo C. the Moon D. Bengaluru

**27. PSLV launched India's Mars Orbiter Mission towards**

A. Mars B. Jupiter C. the Sun D. Neptune

**28. Aditya-L1 is India's first dedicated space mission to observe the**

A. Moon B. Sun C. Earth's oceans only D. asteroid belt

**29. Gaganyaan spacecraft systems are being developed to carry**

A. only weather balloons B. comets C. Indian astronauts D. ocean ships

**30. A space suit provides an astronaut with air and**

A. a natural atmosphere B. a Moon crater C. a new planet D. protection from the space environment

**31. Mars rovers are controlled using signals sent from**

A. Earth B. the Sun's core C. inside Jupiter D. the Milky Way centre

**32. JAXA is the space agency of**

A. India B. Japan C. France only D. Brazil only

**33. Newton's third law explains that forces come in**

A. circles only B. different colours C. equal and opposite pairs D. units of time

**34. An object moving faster has more kinetic energy if its mass is**

A. zero B. unknown only C. always smaller D. the same

**35. Heat naturally flows from a hotter object to a**

A. colder object B. hotter object only C. star only D. vacuum only

**36. An electric circuit needs a complete path for**

A. gravity to stop B. current to flow C. light to vanish D. sound to freeze

**37. Which number should replace the blank? 5, 10, 15, __, 25**

A. 18 B. 30 C. 20 D. 35

**38. A table shows temperatures of 20°C, 25°C, 22°C and 25° C. The highest value occurs**

A. once B. three times C. four times D. twice

**39. The Chandrayaan-3 landing place was in the Moon's**

A. southern high-latitude region B. equatorial ocean C. atmosphere D. north pole of Earth

**40. A modern application of satellites is**

A. making stars B. tracking cyclones C. stopping seasons D. turning planets into moons

## OLYMPIAD CHALLENGE - QUESTIONS 41-50

*These ten questions are deliberately more demanding. Read carefully, reason step by step and calculate where needed.*

**41. A star is 20° above the northern horizon. Which pair describes it correctly?**

A. altitude north and direction 20° B. altitude 70° and direction south C. altitude 20° and direction north D. distance 20 km and direction north

**42. Which order is correct from the Sun outward?**

A. Venus, Mercury, Mars, Earth B. Earth, Venus, Mercury, Mars C. Mercury, Earth, Venus, Mars D. Mercury, Venus, Earth, Mars

**43. Object A is a cloud of gas and dust. Object B is a huge group of stars. A and B are respectively**

A. nebula and galaxy B. galaxy and constellation C. star cluster and planet D. meteor and nebula

**44. If it is 12 noon at longitude 0°, the basic time-zone rule gives about 2 p.m. at 30° east. Why?**

A. Earth stops at noon B. Earth rotates about 15° per hour C. the Moon adds two hours D. east always means midnight

**45. A satellite image shows less green area this year than last year. The most reasonable first conclusion is that**

A. gravity has stopped B. the satellite made the land brown C. vegetation cover may have decreased D. all plants on Earth are gone

**46. A rocket produces 800 units of thrust and has 300 units of opposing force. The unbalanced upward force is**

A. 1100 units B. 300 units C. 240000 units D. 500 units

**47. Which statement is accurate in 2026?**

A. Gaganyaan is undergoing development and qualification tests B. Gaganyaan has already landed people on Mars C. Aditya-L1 studies Jupiter D. Chandrayaan-3 was a weather satellite

**48. A message from Earth takes several minutes to reach a Mars rover. This means the rover must**

A. wait for a person beside it B. perform some actions automatically C. travel faster than light D. use sound through space

**49. A 2 kg object moves at 3 m/s. Using momentum = mass x speed, its momentum is**

A. 1.5 kg m/s B. 5 kg m/s C. 6 kg m/s D. 9 kg m/s

**50. A student changes both the lamp distance and screen material while testing shadow size. Why is the test weak?**

A. the lamp gives light B. shadows can be measured C. the screen has a surface D. more than one variable was changed

# MOCK PAPER 4

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes     Maximum Marks: 50

*Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question.*

**1. The line where the sky seems to meet the ground is the**

A. zenith B. orbit C. equator only D. horizon

**2. The point directly overhead in the sky is called the**

A. zenith B. nadir C. horizon D. eclipse

**3. Which is a good condition for observing faint stars?**

A. a brightly lit stadium B. a dark place away from city lights C. looking through thick clouds D. standing under a streetlamp

**4. A meteor shower is caused when Earth passes through**

A. a star cluster B. the Sun’s surface C. a stream of small particles D. a rocket launch pad

**5. Which planet has the hottest average surface because of its thick atmosphere?**

A. Mercury B. Mars C. Neptune D. Venus

**6. Uranus and Neptune are often called**

A. ice giants B. terrestrial planets C. dwarf stars D. natural satellites

**7. A natural satellite moves around a**

A. constellation B. planet or dwarf planet C. galaxy only D. rocket stage

**8. The asteroid belt is a region containing many rocky bodies orbiting the**

A. Moon B. Earth only C. Sun D. Milky Way centre

**9. A meteoroid is found mainly**

A. only on Earth’s ground B. inside a telescope C. inside a star D. in space before entering an atmosphere

**10. Which property tells how hot a star’s surface is most directly?**

A. colour B. number of planets only C. constellation name D. distance from school

**11. A group of stars born from the same cloud may form a**

A. time zone B. star cluster C. launch fairing D. meteorite field only

**12. The Milky Way appears as a faint band because we are viewing many distant**

A. oceans B. rockets C. stars D. Moon phases

**13. Andromeda is another**

A. planet B. constellation only C. rocket engine D. galaxy

**14. The layer of gases around Earth is the**

A. atmosphere B. crust C. mantle D. core

**15. When the Northern Hemisphere is tilted away from the Sun, it receives**

A. more direct sunlight B. less direct sunlight C. no sunlight anywhere D. light only from the Moon

**16. A first-quarter Moon appears approximately**

A. fully dark B. fully lit C. half lit D. as a solar eclipse

**17. Ocean tides are caused mainly by the gravity of the**

A. North Star B. asteroid belt C. International Space Station D. Moon

**18. A weather satellite usually carries instruments that observe clouds and**

A. temperature patterns B. star birth only C. rocket paint D. Moon rocks only

**19. A navigation receiver uses signals from several satellites to calculate its**

A. mass B. position C. colour D. temperature of the Sun

**20. An Earth-observation satellite can help map damage after a**

A. constellation change B. solar-system formation C. cyclone D. star cluster birth

**21. The main body of a spacecraft that holds systems and instruments is called the**

A. meteor train B. launch shadow C. planet crust D. spacecraft bus

**22. The engine nozzle directs hot gases mainly**

A. backward B. towards the payload C. into the fuel tank D. straight into the Sun

**23. A rocket stage includes engines, fuel tanks and**

A. oceans B. supporting structures C. constellations D. Moon phases

**24. The payload is separated from the launch vehicle after reaching the required**

A. classroom B. continent C. path or orbit D. star colour

**25. A reusable booster may return by landing or**

A. becoming a comet B. entering the Sun C. turning into a planet D. splashing down for recovery

**26. Vikram Sarabhai is widely known as a key founder of India's**

A. space programme B. ocean programme C. railway network D. weather seasons

**27. GSLV uses powerful stages to place satellites into**

A. the ocean floor B. higher-energy orbits C. Moon craters without launch D. inside Earth's core

**28. Chandrayaan-2 included an orbiter designed to study the**

A. Sun B. Mars C. Moon D. Saturn

**29. The aim of Gaganyaan includes sending Indian crew to**

A. the surface of the Sun B. another galaxy C. inside Jupiter D. low Earth orbit

**30. Astronauts train in water partly because it can help practise**

A. movement during spacewalk-like tasks B. planet formation C. solar eclipses D. building stars

**31. A Moon mission may use an orbiter, lander and**

A. tide B. rover C. constellation D. time zone

**32. Roscosmos is the space agency of**

A. India B. Japan C. Russia D. Canada only

**33. A moving object remains in motion unless an unbalanced force acts. This describes**

A. Newton's third law only B. reflection C. conduction D. Newton's first law

**34. Potential energy is energy stored because of position or**

A. condition B. colour only C. name D. sound level only

**35. A mirror changes the direction of light by**

A. conduction B. reflection C. evaporation D. gravity

**36. An electromagnet works when electric current flows through a**

A. wooden block B. glass of water only C. coil of wire D. shadow

**37. Which fraction is largest?**

A. 1/4 B. 1/2 C. 2/5 D. 3/4

**38. A student measures a shadow at 9 a.m., noon and 3 p.m. The shadow is usually shortest near**

A. noon B. 9 a.m. only C. 3 p.m. only D. midnight

**39. Aditya-L1 scientific data help researchers study solar activity and**

A. ocean tides only B. space weather C. Earth's core D. meteorite names

**40. One benefit of space technology during disasters is**

A. stopping all storms B. removing gravity C. rapid communication and mapping D. changing Earth's orbit

## OLYMPIAD CHALLENGE - QUESTIONS 41-50

*These ten questions are deliberately more demanding. Read carefully, reason step by step and calculate where needed.*

**41. A student sees a star exactly halfway between the horizon and the zenith. Its altitude is about**

A. 0° B. 90° C. 180° D. 45°

**42. A small body orbits the Sun, is mostly ice and forms a tail near the Sun. It is a**

A. comet B. asteroid only C. moon D. dwarf star

**43. Which order goes from smaller system to larger system?**

A. Universe, galaxy, star cluster B. star cluster, galaxy, Universe C. galaxy, Universe, star cluster D. star cluster, Universe, galaxy

**44. A place at 45° east is about three hours ahead of 0° longitude because**

A. the Moon moves 15° each hour B. seasons add hours C. Earth rotates about 15° each hour D. eclipses move clocks

**45. Satellite A stays above nearly the same region; Satellite B passes over many latitudes. A and B are most likely**

A. both natural satellites B. both fixed on the ground C. polar-orbiting and launch vehicle D. geostationary-type and polar-orbiting

**46. A rocket has three stages. Which stage must usually fire first?**

A. the lowest stage at lift-off B. the payload C. the top stage after landing D. all stages only after orbit

**47. Which mission status is correctly stated?**

A. Chandrayaan-3 is orbiting Mars B. Aditya-L1 is operating as a solar observatory C. Gaganyaan has landed humans on the Moon D. PSLV is a space station

**48. A rover receives a command 10 minutes after it is sent. An immediate hazard appears before the command arrives. The rover needs**

A. a louder speaker B. a shorter planet C. onboard hazard-detection software D. sound waves through vacuum

**49. A 4 kg object speeds up from 1 m/s to 3 m/s. Its momentum increases by**

A. 2 kg m/s B. 4 kg m/s C. 12 kg m/s D. 8 kg m/s

**50. A plant experiment uses identical pots but different amounts of water. What should be kept the same?**

A. plant type, soil, light and time B. water amount C. final height only D. result

# MOCK PAPER 5

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes     Maximum Marks: 50

*Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for each question.*

**1. North, south, east and west are called the four main**

A. directions  B. phases  C. orbits  D. galaxies

**2. A star chart is used to help identify**

A. rocket fuel  B. stars and constellations  C. Earth's core  D. satellite batteries

**3. Binoculars can help beginners observe the Moon because they**

A. create moonlight  B. stop clouds  C. magnify distant objects  D. change gravity

**4. The best reason to repeat a sky observation on several nights is to**

A. make the sky move faster  B. increase the number of planets  C. change the weather  D. notice patterns and changes

**5. Which planet is known for having a Great Red Spot?**

A. Jupiter  B. Venus  C. Mercury  D. Earth

**6. Which planet is farthest from the Sun among the eight planets?**

A. Mars  B. Neptune  C. Saturn  D. Uranus

**7. Earth's Moon is a**

A. dwarf planet  B. star  C. natural satellite  D. comet

**8. Ceres is found mainly in the**

A. Sun's atmosphere  B. Kuiper Belt beyond Neptune only  C. Earth's ocean  D. asteroid belt

**9. A meteor is commonly called a**

A. shooting star  B. new planet  C. space station  D. Moon phase

**10. A red star is usually cooler than a**

A. red star of the same type  B. blue star  C. planet  D. meteorite

**11. Stars in a constellation may look close together but can actually be at very different**

A. temperatures only  B. time zones  C. distances  D. rocket stages

**12. The Milky Way is only one of many**

A. moons  B. planets  C. launch sites  D. galaxies

**13. The space between galaxies is part of the**

A. Universe  B. Earth's crust  C. rocket fairing  D. troposphere only

**14. The inner core of Earth is mainly solid because of very high**

A. rainfall  B. pressure  C. moonlight  D. wind speed

**15. Earth's atmosphere contains the air needed by**

A. stars  B. asteroids  C. living things  D. rocket stages only

**16. The Moon takes about one month to go around**

A. the Sun directly only  B. Mars  C. Jupiter  D. Earth

**17. A solar eclipse happens when the Moon blocks part or all of the**

A. Sun B. Earth C. Milky Way D. north direction

**18. A satellite that studies crops and forests is an**

A. natural satellite B. Earth-observation satellite C. astronaut D. launch pad

**19. A communication satellite often receives, strengthens and sends**

A. gravity B. tides C. signals D. meteorites

**20. Weather forecasts improve when satellites provide pictures of**

A. star clusters B. rocket stages C. Moon rocks only D. cloud movement

**21. A satellite needs a power system, communication system and**

A. control system B. ocean system C. constellation system D. tide system

**22. A rocket must carry its own oxidiser because space has almost no**

A. gravity B. air C. light D. temperature

**23. The first few seconds after launch require strong thrust to overcome gravity and**

A. Moon phases B. time zones C. air resistance D. star colour

**24. The final rocket stage may place a satellite into the correct**

A. continent B. ocean C. constellation D. orbit

**25. A payload can be a satellite, spacecraft or scientific**

A. instrument B. season C. shadow only D. time zone

**26. Aryabhata was India's first**

A. crew member B. satellite C. Moon rover D. launch site

**27. The Chandrayaan-3 rover was named**

A. Vikram B. Rohini C. Pragyan D. Bhaskara

**28. The Chandrayaan-3 lander was named**

A. Pragyan B. Mangalyaan C. Aryabhata D. Vikram

**29. Aditya-L1 was launched using a**

A. PSLV B. submarine C. weather balloon D. Moon rover

**30. An astronaut outside a spacecraft wears a suit that supplies**

A. soil B. oxygen C. rain D. natural gravity

**31. Space probes can visit worlds that are too distant for**

A. radio signals B. light C. human crews at present D. gravity

**32. The China National Space Administration is commonly shortened to**

A. ISRO B. NASA C. JAXA D. CNSA

**33. Friction is a force that usually opposes**

A. motion B. mass C. time D. temperature

**34. Energy stored in a stretched rubber band is**

A. sound only B. potential energy C. light only D. nuclear energy only

**35. White light can split into colours when it passes through a**

A. metal block B. magnet C. prism D. battery only

**36. Two like poles of magnets**

A. attract each other B. produce gravity only C. become light D. repel each other

**37. Which shape comes next: circle, square, circle, square, __?**

A. circle B. triangle C. rectangle D. star only

**38. A graph rises from 2 to 5 and then to 8. The increase each step is**

A. 2 B. 3 C. 5 D. 8

**39. In 2026, Gaganyaan was still undergoing important**

A. crewed landings on Mars B. construction on the Sun C. tests and qualification work D. flights to another galaxy

**40. Satellites help fishermen and sailors through weather information and**

A. Moon phases only B. star formation C. rocket thrust D. navigation

## OLYMPIAD CHALLENGE - QUESTIONS 41-50

*These ten questions are deliberately more demanding. Read carefully, reason step by step and calculate where needed.*

**41. An observer turns from north through a quarter turn clockwise. The observer now faces**

A. east B. west C. south D. north-west

**42. Planet A is rocky and close to the Sun. Planet B is very large and has many moons. A and B are most likely**

A. two dwarf planets B. terrestrial planet and giant planet C. two stars D. comet and meteor

**43. If a star looks faint, which conclusion is safest?**

A. it must be the smallest star B. it must be outside the Universe C. it may be far away, small, cool, or a combination D. it gives no light

**44. The Moon is full tonight. About one week later it will be near the**

A. new Moon immediately B. first-quarter phase C. solar eclipse every time D. third-quarter phase

**45. A phone finds its position using signals from four navigation satellites. Why are several satellites useful?**

A. they allow the receiver to compare signal travel information B. one satellite creates all gravity C. they make the phone heavier D. they stop Earth rotating

**46. A rocket's total upward thrust is 1000 N while gravity and drag together act downward with 700 N. The net force is**

A. 1700 N upward B. 300 N upward C. 300 N downward D. 700 N upward

**47. Which pair is correctly matched?**

A. Vikram - Mars Orbiter B. Aditya-L1 - lunar rover C. Pragyan - Chandrayaan-3 rover D. PSLV - space station

**48. Why can a spacecraft not use ordinary sound to talk directly through empty space?**

A. sound is too bright B. space has too much water C. gravity blocks all sound equipment D. sound needs a material medium

**49. A battery lights two identical bulbs in series. If one bulb is removed and the circuit breaks, the other bulb goes out because**

A. the path for current is incomplete B. gravity is removed C. light cannot reflect D. the battery becomes a magnet only

**50. A student records rocket-model distances of 6 m, 8 m, 8 m, 9 m and 11 m. The mode is**

A. 6 m B. 8 m C. 9 m D. 11 m

# ANSWER KEYS

## Mock Paper 1

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A |
| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B |

## Mock Paper 2

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B |
| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C |

## Mock Paper 3

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C |
| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |

## Mock Paper 4

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |
| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A |

## Mock Paper 5

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A |
| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B |

# OLYMPIAD CHALLENGE - ANSWER EXPLANATIONS

## Mock Paper 1

**41. A -** East and west are opposite directions, so west is directly behind a person facing east.

**42. B -** The planetary order begins Mercury, Venus, Earth, Mars.

**43. C -** Stars produce their own light, while planets and moons mainly reflect light.

**44. D -** Moving one time zone east usually adds one hour.

**45. A -** Remote sensing instruments collect information about Earth from a distance.

**46. B -** Dropping an empty stage reduces the mass that later stages must accelerate.

**47. C -** Chandrayaan is India's lunar programme.

**48. D -** The station and astronauts continuously fall around Earth in orbit, producing apparent weightlessness.

**49. A -** Work = force x distance = 5 x 4 = 20 joules.

**50. B -** The repeated value is 12, recorded on the second and third nights.

## Mock Paper 2

**41. B -** A useful observation records when and where the object was seen and its position in the sky.

**42. C -** A round body orbiting the Sun that has not cleared its orbital region may be classified as a dwarf planet.

**43. D -** A galaxy can contain billions of stars along with clusters and nebulae.

**44. A -** Seasonal heating depends mainly on Earth's axial tilt and the angle of sunlight.

**45. B -** Earth-observation satellites revisit regions and collect comparable images.

**46. C -** The empty stage adds mass, so later engines must accelerate more unnecessary material.

**47. D -** PSLV and GSLV are launch-vehicle families designed for different payloads and orbits.

**48. A -** Long stays in microgravity can weaken muscles and bones, so exercise is essential.

**49. B -** Work = force x distance = 12 x 3 = 36 joules.

**50. C -** Each number is twice the previous number.

## Mock Paper 3

**41. C -** Altitude is the angle above the horizon; north gives the compass direction.

**42. D -** The first four planets are Mercury, Venus, Earth and Mars.

**43. A -** A nebula is a gas-and-dust cloud; a galaxy is a vast system of stars and other matter.

**44. B -** Earth turns 360° in 24 hours, or about 15° each hour.

**45. C -** A change in green area can indicate vegetation change, but further evidence is needed before a stronger claim.

**46. D -** Net upward force = thrust - opposing force = 800 - 300 = 500.

**47. A -** ISRO continued Gaganyaan qualification and parachute-related tests in 2026.

**48. B -** Because of communication delay, rovers need onboard software for safe and partly autonomous operation.

**49. C -** Momentum = 2 x 3 = 6 kg m/s.

**50. D -** A fair test changes one factor at a time while keeping other conditions the same.

## Mock Paper 4

**41. D -** The angle from horizon to zenith is 90°, so halfway is about 45°.

**42. A -** Icy bodies that form tails near the Sun are comets.

**43. B -** A star cluster is within a galaxy, and galaxies are part of the Universe.

**44. C -** 45° divided by 15° per hour gives about 3 hours.

**45. D -** A geostationary satellite appears fixed over one longitude; a polar satellite covers many latitudes.

**46. A -** The lowest or first stage provides initial thrust during lift-off.

**47. B -** Aditya-L1 continues solar observations from near the Sun-Earth L1 point.

**48. C -** Communication delay makes onboard sensing and autonomous safety actions necessary.

**49. D -** Change in momentum = m(v2-v1) = 4(3-1) = 8 kg m/s.

**50. A -** Only the amount of water should change; other important conditions should remain controlled.

## Mock Paper 5

**41. A -** A quarter turn clockwise from north points east.

**42. B -** Rocky inner planets are terrestrial; very large outer planets are giants.

**43. C -** Apparent brightness depends on several factors, especially luminosity and distance.

**44. D -** The main phases are about one week apart: full Moon is followed by third quarter.

**45. A -** Position is found by comparing timing and distance information from multiple satellites.

**46. B -** Net force = 1000 - 700 = 300 N upward.

**47. C -** Pragyan was the Chandrayaan-3 rover; Vikram was the lander.

**48. D -** Sound waves require matter such as air, while radio waves can travel through vacuum.

**49. A -** Electric current needs a complete conducting path.

**50. B -** The mode is the value that occurs most often; 8 m appears twice.

# SYLLABUS COVERAGE AUDIT

| Unit | Paper 1 | Paper 2 | Paper 3 | Paper 4 | Paper 5 |
|---|---|---|---|---|---|
| Fundamentals of Astronomy | 5 | 5 | 5 | 5 | 5 |
| The Solar System | 6 | 6 | 6 | 6 | 6 |
| Stars, Galaxies and the Universe | 5 | 5 | 5 | 5 | 5 |
| Earth and Moon | 5 | 5 | 5 | 5 | 5 |
| Satellites and Space Technology | 5 | 5 | 5 | 5 | 5 |
| Rockets and Launch Vehicles | 5 | 5 | 5 | 5 | 5 |
| ISRO and India's Space Missions | 5 | 5 | 5 | 5 | 5 |

| | | | | | |
|---|---|---|---|---|---|
| Space Exploration | 4 | 4 | 4 | 4 | 4 |
| Basic Physics for Space Science | 5 | 5 | 5 | 5 | 5 |
| Scientific Reasoning and Logical Aptitude | 3 | 3 | 3 | 3 | 3 |
| General Space Awareness | 2 | 2 | 2 | 2 | 2 |

## Difficulty and Scoring Guide

Questions 1-15 are mainly direct recall and recognition. Questions 16-40 combine understanding and application. Questions 41-50 are the Olympiad Challenge and include multi-step reasoning or short calculations.

| Score | Suggested interpretation |
|---|---|
| 45-50 | Excellent Olympiad readiness |
| 38-44 | Strong preparation; review challenge errors |
| 28-37 | Good foundation; revise weaker units |
| 18-27 | Developing understanding; practise unit by unit |

| | |
|---|---|
| 0-17 | Revisit the Junior Level reading material before retesting |